KI Service Zentrum
by Hasso-Plattner-Institut

HPI

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Efficient Retrieval of Information

Local Retrieval Augmented-Generation

Hanno Müller

Design IT.
Create Knowledge.

www.hpi.de

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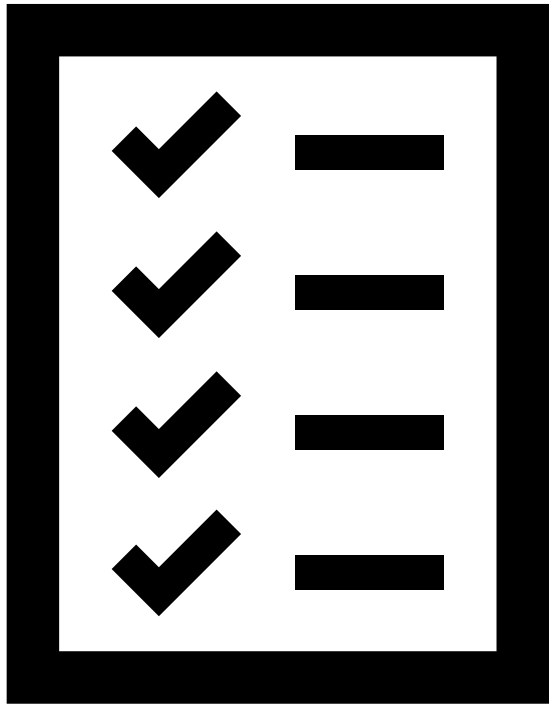
Attendencee list



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LEARNING GOALS

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Motivation

- LLMs are trained using (mostly) public data. Thus, LLMs have no knowledge of **private data**, e.g.
  - **internal documents** about operational processes
  - **paid documents** such as scientific publications
- LLMs mix data. If **no specific information** is available to answer a question, similar information is used, e.g.
  - legal texts on parking restrictions are used to answer questions about stopping restrictions
  - popular science texts are used to answer research questions
- RAG allows **predefined documents** to be used as a basis for answering questions!

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Learnings

- Understand the fundamentals of RAG
  - Embeddings
  - Vector databases
  - Information retrieval
  - Prompt engineering
- Increase awareness for the requirements for (your own) RAG
  - e.g., document preprocessing
  - e.g., creating vector databases
  - e.g., prompt engineering
- Deploy a locally running RAG application on your machine

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Agenda

1. General introduction (30 minutes)
2. Demonstration of RAG (30 minutes)
3. Interactive Notebook (30 minutes)
4. Deploy RAG + Break (30 minutes)
5. Customise RAG (30 minutes)
6. Discussion (30 minutes)

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FUNDAMENTALS

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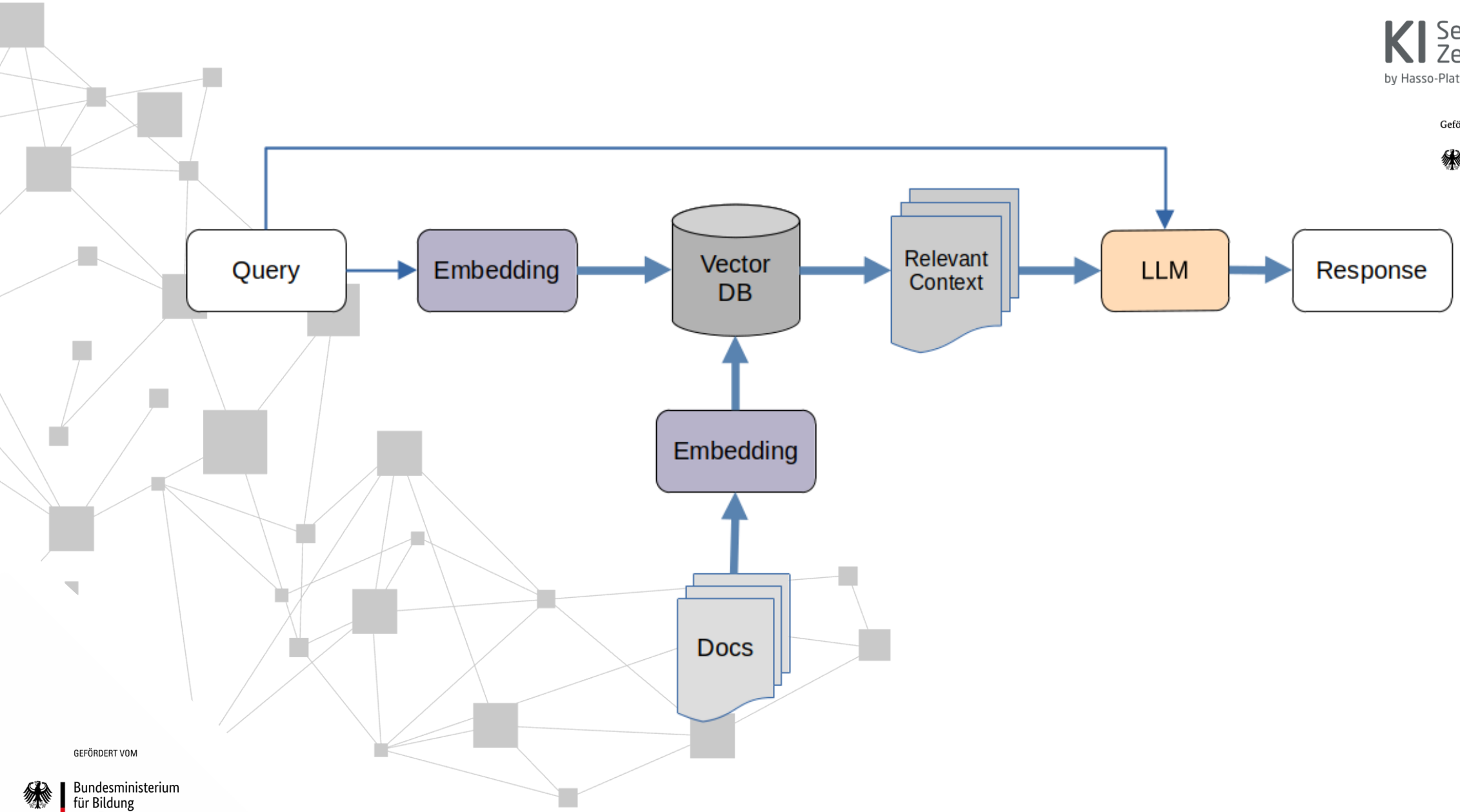

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GEFÖRDERT VOM
Bundesministerium für Bildung und Forschung

Local RAG Tool

RAG Query | Semantic Retrieval | RAG Chat | Chat | Rag Tool Settings

Query an LLM about information from your documents.

Query

Settings

Submit

Answer

Export

Hands-On

https://github.com/aihpi/workshop-rag/blob/main/learning-materials/01_workflow.ipynb

https://github.com/aihpi/

→ repositories

→ workshop-rag

→ learning-materials

→ 01_workflow.ipynb

GEFÖRDERT VOM

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EXERCISE

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Exercise

1. Upload your own documents to the vector database.

2. Adjust the prompt so that answers are bullet point lists, e.g.

   Questions:
   Who is Watson?

   Answer:
   - A fictional character from Sherlock Holmes stories
   - Holmes' friend
   - First appeared in "A Study in Scarlet" (1887)

3. Optimise relevance threshold, top K, and temperature for improved responses. What do these parameters do?

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DISCUSSION

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Four AI Service Centres in Germany

KI Service Zentrum by Hasso-Plattner-Institut

https://hpi.de/ki-servicezentrum/

Overview

KI-Servicezentren

Goal:
Reduce barriers to the implementation of AI applications in society and the economy

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AI Service Centre Berlin-Brandenburg

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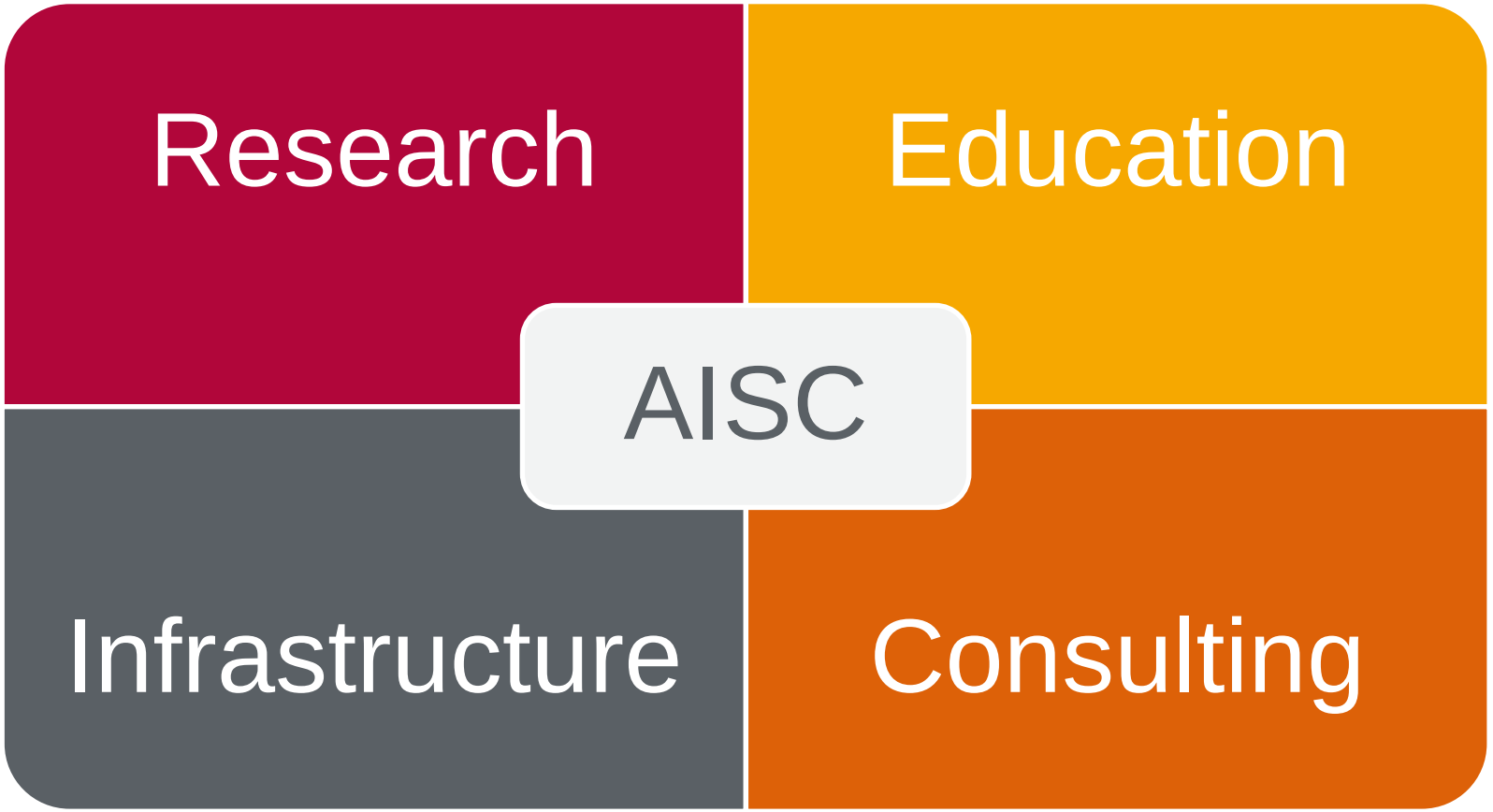
Structure



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Research

- **AI methods research**
- **AI operations research**

PubMedCLIP: How Much Does CLIP Benefit Visual Question Answering in the Medical Domain?

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Abstract

Contrastive Language–Image Pre-training (CLIP) has shown remarkable success in learning with cross-modal supervision from extensive amounts of image–text pairs collected online. Thus far, the effectiveness of CLIP has been investigated primarily in general-domain multimodal problems. In this work, we evaluate the effectiveness of CLIP for the task of Medical Visual Question Answering (MedVQA). We present PubMedCLIP, a fine-tuned version of CLIP for the medical domain based on PubMed articles. Our experiments conducted on two MedVQA benchmark datasets illustrate that PubMedCLIP achieves superior results improving the overall accuracy up to 3% in comparison to the state-of-the-art Model-Agnostic Meta-Learning (MAML) networks pre-trained only on visual data. The PubMedCLIP model with different back-ends, the source code for pre-training them and reproducing our MedVQA pipeline is publicly available at https://github.com/sarahESL/PubMedCLIP.

1 Introduction

Medical visual question answering (MedVQA) seeks answers to natural language questions about a given medical image. The development of MedVQA has considerable potential to benefit healthcare systems, as it may aid clinicians in interpreting medical images and obtaining more accurate diagnoses by consulting a second opinion. Thus, it has become a very active area of research, with competitive benchmarks and yearly competitions (Abacha et al., 2021). Yet, visual question answering in the medical domain in particular remains non-trivial as we suffer from a general lack of large balanced training data, in part due to privacy concerns. To solve the multimodal task of MedVQA, a system must understand both medical images and textual questions and infer the associations between them sufficiently well to produce a correct answer (An

Findings of the Association for Computati
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Exploring Paracrawl for Document-level Neural Machine Translation

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Abstract

Document-level neural machine translation (NMT) has outperformed sentence-level NMT on a number of datasets. However, document-level NMT is still not widely adopted in real-world translation systems mainly due to the lack of large-scale general-domain training data for document-level NMT. We examine the effectiveness of using Paracrawl for learning document-level translation. Paracrawl is a large-scale parallel corpus crawled from the Internet and contains data from various domains. The official Paracrawl corpus was released as parallel sentences (extracted from parallel webpages) and therefore previous works only used Paracrawl for learning sentence-level translation. In this work, we extract parallel paragraphs from Paracrawl parallel webpages using automatic sentence alignments and we use the extracted parallel paragraphs as parallel documents for training document-level translation models. We show that document-level NMT models trained with only parallel paragraphs from Paracrawl can be used to translate real documents from TED, News and Europarl, outperforming sentence-level NMT models. We also perform a targeted pronoun evaluation and show that document-level models trained with Paracrawl data can help context-aware pronoun translation. We release our data and code here[1].

1 Introduction

The Transformer translation model (Vaswani et al., 2017), which performs sentence-level translation based on attention networks, has achieved great success and significantly improved the state-of-the-art in machine translation. Compared to sentence-level translation, document-level translation (Xu et al., 2021; Bao et al., 2021; Jauregi Unanue et al., 2020; Ma et al., 2020; Maruf et al., 2019; Tu et al., 2018; Maruf and Haffari, 2018) performs translation at document-level and can potentially fur-

[1]https://github.com/Yusser96/Exploring-Paracrawl-for-Document-level-Neural-Machine-Translation

Proceedings of the 17th Conference of the European Chapter of
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Efficient Parallelization Layouts for Large-Scale Distributed Model Training

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Abstract

Efficiently training large language models requires parallelizing across hundreds of hardware accelerators and invoking various compute and memory optimizations. When combined, many of these strategies have complex interactions regarding the final training efficiency. Prior work tackling this problem did not have access to the latest set of optimizations, such as FlashAttention or sequence parallelism. In this work, we conduct a comprehensive ablation study of possible training configurations for large language models. We distill this large study into several key recommendations for the most efficient training. For instance, we find that using a micro-batch size of 1 usually enables the most efficient training layouts. Larger micro-batch sizes necessitate activation checkpointing or higher degrees of model parallelism and also lead to larger pipeline bubbles. Our most efficient configurations enable us to achieve state-of-the-art training efficiency results over a range of model sizes, most notably a Model FLOPs utilization of 70.5% when training a Llama 13B model.

1 Introduction

The number of parameters and computational resources spent on training deep neural networks is growing rapidly [1, 3, 14]. The largest models consisting of hundreds of billions of parameters do not even fit onto a single hardware accelerator. Thus, training these models requires various ways of reducing the memory requirements, such as ZeRO [16], activation checkpointing [2], and 3D-parallel (data, tensor, and pipeline parallel) training [13]. 3D parallelism, in particular, has been demonstrated to be effective for the training of Transformer-based large language models (LLMs) with hundreds of billions of parameters [13].

However, training these models efficiently with 3D parallelism requires significant domain expertise and extensive manual effort to determine the ideal configurations. These configurations not only need to combine data, model, and pipeline parallelism most efficiently, but also consider complex interactions with other memory and compute optimizations. FlashAttention [5] in particular has had a notable impact since its release, enabling us to train models at previously impossible degrees of training efficiency. In light of these developments, we conduct a systematic study via a large-scale training efficiency sweep of these interactions. We consider up to 256 GPUs and Llama models with up to 65 billion parameters.

Workshop on Advancing Neural Network Training at 37th Conference on Neural Information Processing Systems (WANT@NeurIPS 2023).

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Zugangsanfrage
aisc.hpi.de

Infrastructure

- **Free access**
- No production operation
  - Data should be **anonymised** or **synthesised**
  - No **hosting** of products
- **Reporting** & **publication** by users
- **Old rights** remain with users
- **New rights** remain with users
  - Granting of rights of use for research and teaching

Training
- 64 NVIDIA H100 GPU

Inference
- 40 NVIDIA A30 GPU

ARM Server
- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

GPU Server
- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

Edge
- ARMv8 CPU
- NVIDIA Jetson AGX Module

Neuromorph
- 288 SpiNNaker2 Chips

Storage
- 1.5 PB NVMe

Network
- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

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Newsletter

Education

Talks

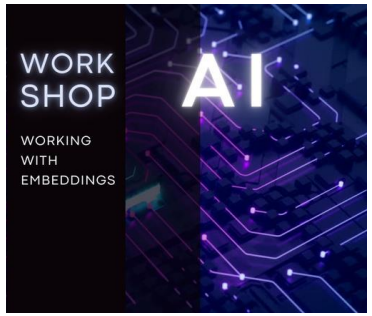
tele-task.de/series/1463

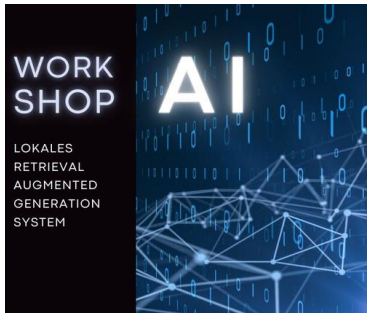
- Guest talks about research and innovation

Work shops

aimaker.community

- Practical topics
- Example topics: Speech2summary, Docker for ML, semantic search

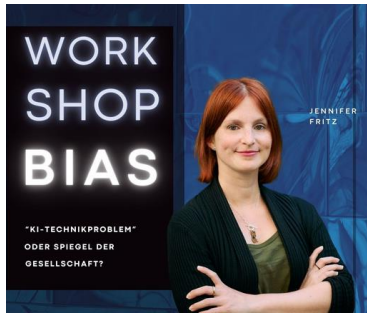


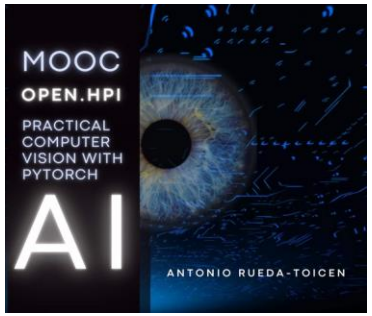


MOOCs

open.hpi.de/channels/ai-service-center

- ChatGPT: What does generative AI mean for our society?
- Profitable AI
- Understanding and avoiding AI biases





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Sprechstunde buchen

Consulting

AI consultation hours

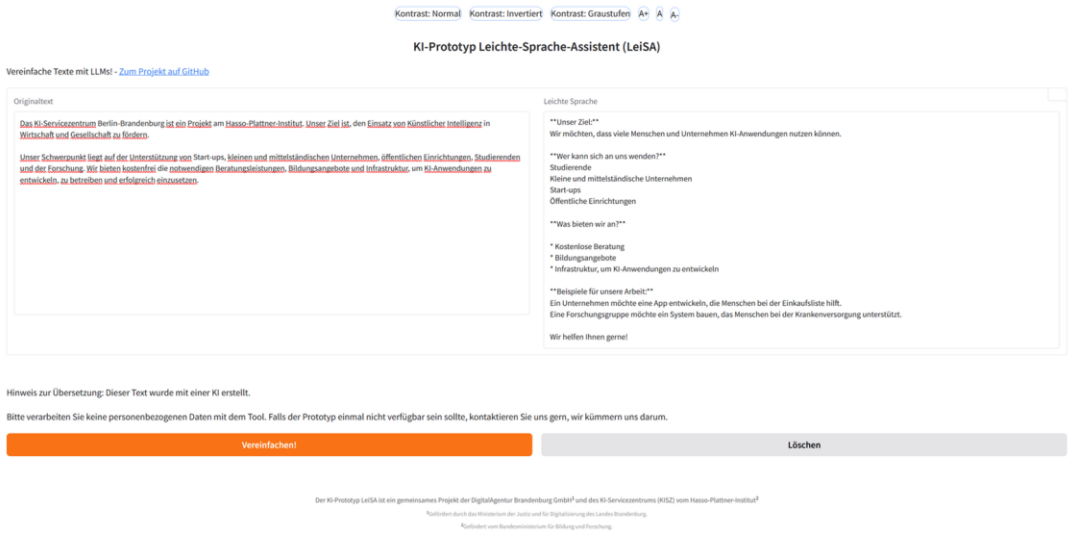
- Answering questions:
  - about AI infrastructure
  - about AI models & frameworks
  - about AI use cases

AI pilot projects

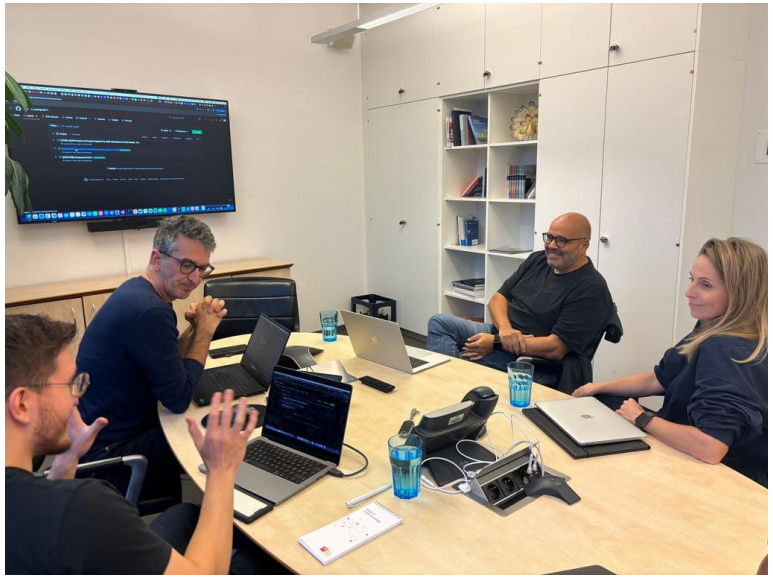
- Co-development of a prototype
- Application every three months
- Selection criteria, e.g. AI maturity, public benefit
- Publication of results

Cooperation

- Jointly organized network meetings



github.com/aihpi/leichte-sprache



Previous AI pilot projects

- Generation of math problems
- Generation of easy language
- Generation of upcycling suggestions
- Reduction of food waste
- Dating by handwriting

Jetzt bewerben!

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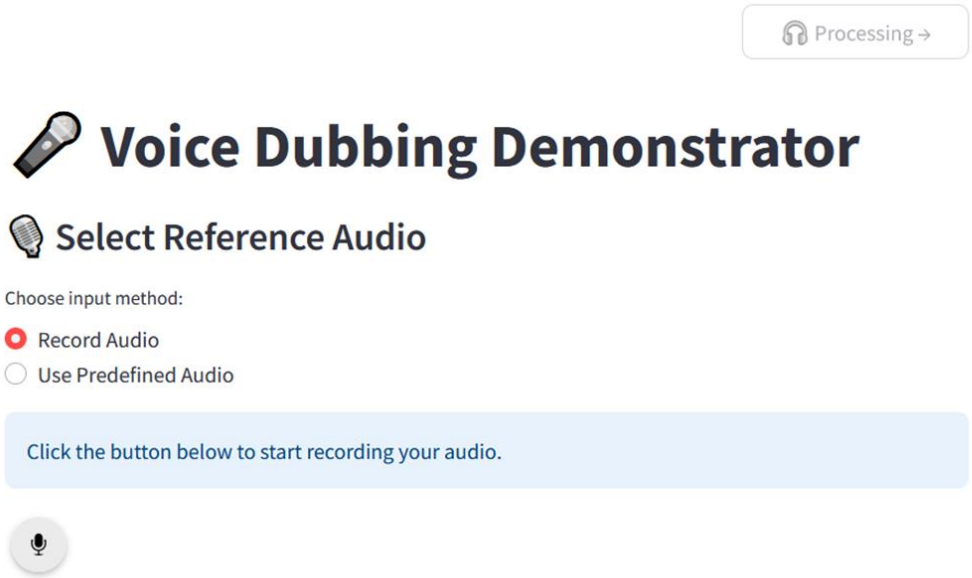
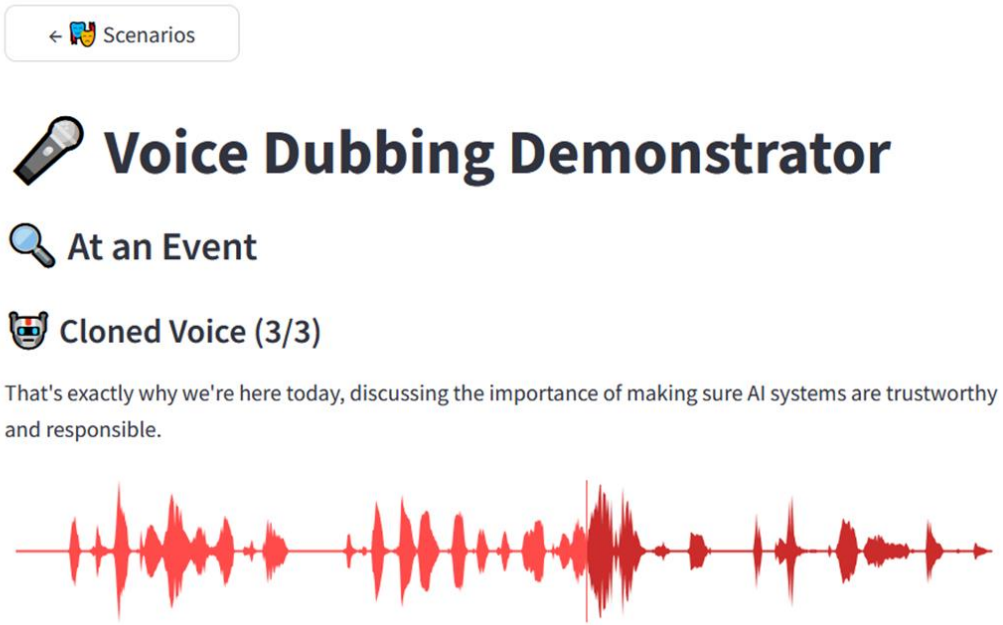
Speech processing

https://tts.aisc.hpi.de/
user: event
pw: aisc@hpi2022!

Prototypes





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